Hyperspectral Imaging and Deep Learning for Rapid Assessment of Soil Nutrient Status Using Ground Robotic Vehicles

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Introduction

Spatio-temporal soil nutrient information is essential for site-specific nutrient management.

Conventional laboratory analysis of nitrogen (N), phosphorus (P), and potassium (K) is slow, costly, and impractical at field scale.

Hyperspectral imaging paired with deep learning enables characterizing soil nutrient status directly from spectral reflectance.

Objectives

Develop a baseline convolutional neural network (CNN) to predict soil N and P content from hyperspectral reflectance.

Evaluate spectral augmentation techniques (SNV, Savitzky–Golay derivatives) to improve model performance.

Apply neural architecture search (NAS) and hyperparameter tuning to optimize the CNN architecture.

Compare model performance on original vs. augmented data to identify the most practical, field-deployable configuration.

Materials & Methods

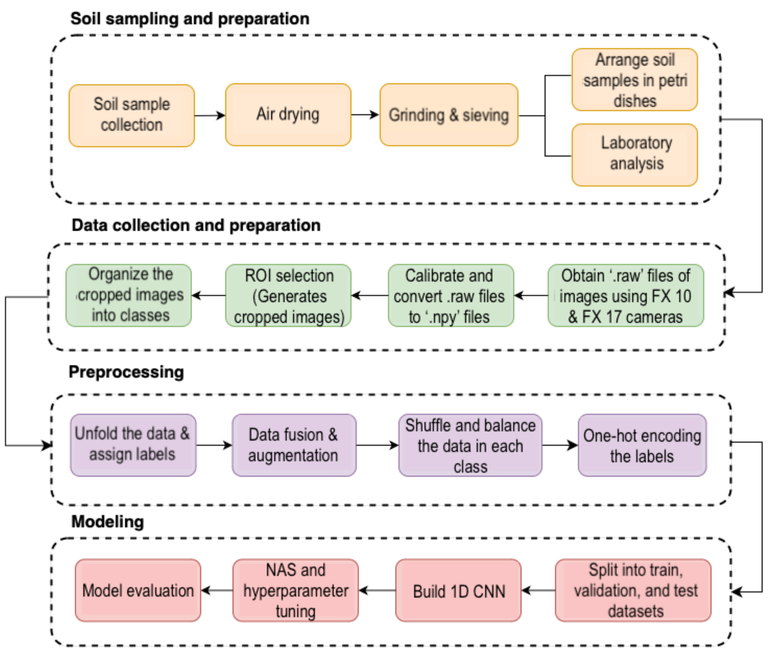


Figure 1. *Flowchart showing methodology of the study*

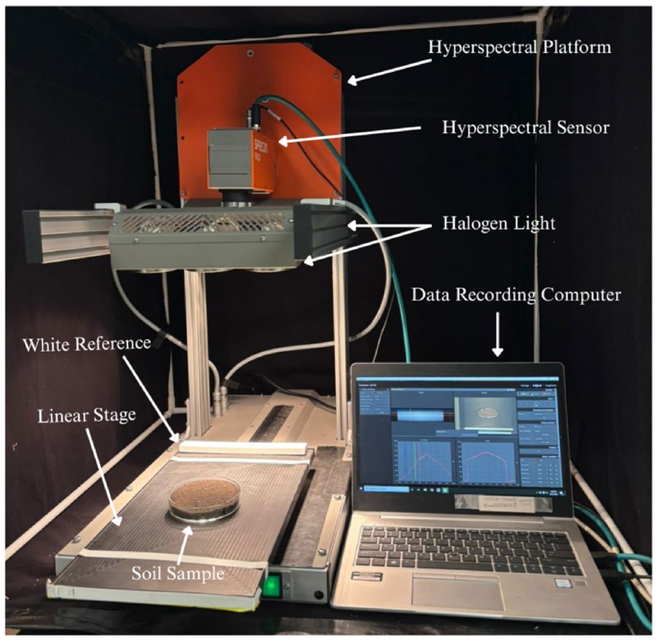


Figure 2. *Hyperspectral data acquisition system*

Results & Discussion

Fusing FX-10 and FX-17 data consistently outperformed either of single sensor for both nutrients (Table 1).

FX-10 (VNIR) alone outperformed FX-17 (SWIR) alone.

Adding SNV (and SNV+SG1 for phosphorus) improved the fused-data baseline modestly (N: 0.67→0.68; P: 0.72→0.74).

NAS improved classification accuracy over the manually tuned baseline for both nutrients, reaching **≈83%** for combined N/P classification (Table 2).

Sensor	N Accuracy	P Accuracy
FX-10	0.47	0.54
FX-17	0.44	0.52
Fused (FX-10+FX-17)	**0.67**	**0.72**

Table 1. *Baseline 1D-CNN accuracy by sensor.*

Model	Acc.	Prec.	Recall	F1
N: CNN-NAS (raw fused)	0.83	0.83	0.83	0.83
N: CNN-NAS (+SNV)	0.83	0.83	0.83	0.83
P: CNN-NAS (raw fused)	0.80	0.80	0.81	0.80
P: CNN-NAS (+SNV+SG1)	0.82	0.82	0.82	0.82

Table 2. *Performance comparison of CNN-NAS models for nitrogen prediction with and without SNV augmentation.*

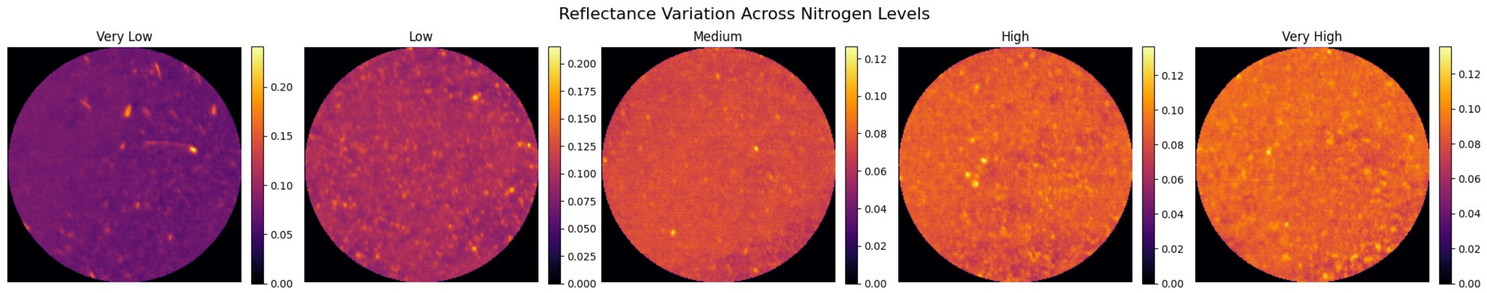
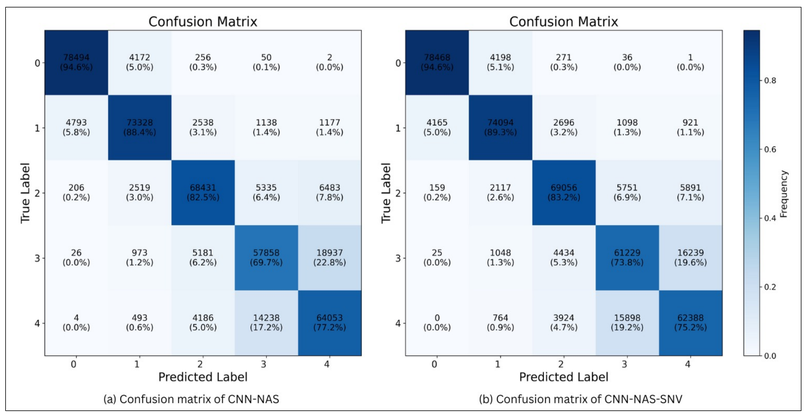
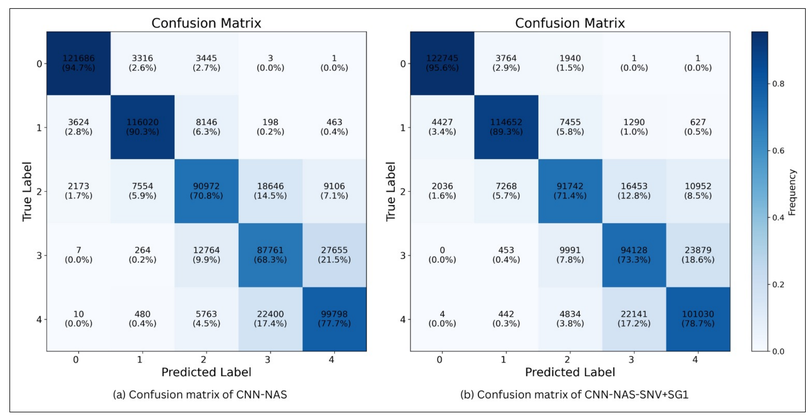


Figure 3. *Variation in spectral reflectance with increasing nitrogen levels*



(1)



(2)

Figure 4. *Confusion matrices for: 1) nitrogen prediction: (a) CNN-NAS on raw fused data, (b) with SNV augmentation, and 2) phosphorus prediction: (a) CNN-NAS on raw fused data, and (b) with SNV+SG1 augmentation.*

Because augmented-data models roughly double the parameter count and training time for only marginal accuracy gains, **CNN-NAS on raw fused data offers the best trade-off for real-time, field-deployable nutrient sensing.**

Conclusions & Future Work

Fused VNIR+SWIR hyperspectral data combined with a NAS-optimized 1D-CNN reliably classifies soil N and P status (≈83% accuracy) in soils.

Spectral augmentation offers only modest accuracy gains at substantial computational cost, favoring simpler models for field deployment.

Proximal hyperspectral sensing can complement conventional soil testing, enabling denser, faster nutrient assessments that better capture field heterogeneity.

Next steps: Feature/band selection, regression models from larger continuous datasets, simultaneous multi-nutrient prediction, and transition to on-the-go, UGV-mounted field sensing.

Acknowledgements & Contact

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Full study: Vullaganti, N., Ram, B.G., Zhang, X. *et al.* AI-Augmented hyperspectral soil sensing: predictive modeling of nitrogen and phosphorus using neural architecture search.*Precision Agric* **27**, 21 (2026). https://doi.org/10.1007/s11119-026-10323-y

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